Centennial College

ETEC326 – Technical Project (SEC. 001), Fall 2026

Milestone 1 – Detailed Design (Preliminary)

Edge–Cloud Collaborative Intelligent Security and Dynamic Target Tracking

(Radar-Cued Pan-Tilt Security Node)

Team member

Tang Li (solo project)

Instructor: [Instructor Name]

Prepared: November 9, 2026

Design stored on GitHub:
https://github.com/Makabakamanihong/ETEC306/tree/main/docs/etec326-milestone1

PRELIMINARY (UNTESTED) DETAILED DESIGN. This milestone records the design before the ETEC326 acceptance tests. Items are labelled VERIFIED, IMPLEMENTED, PRELIMINARY / UNTESTED or PAPER STUDY; untested values may change after the tests planned for November 26 – December 1, 2026.

How to Read This Report

This is the Milestone 1 report for my ETEC326 project. I am the only team member, so the "group" report and my individual work are the same. The report follows the rubric outline: **(1)** emerging project-management decisions on scope, resourcing and scheduling; **(2)** the enabling technologies; **(3)** the preliminary detailed design (hardware and software) with design notes; **(4)** discarded designs, features and aspects with the reasons. Appendix A lists the datasheets that will be posted as an appendix in Milestone 2. My log book is submitted separately.

PRELIMINARY / UNTESTED DESIGN. The design describes software release v6 (deployed on October 20, 2026) and the planned release v7 (feature freeze, November 20). Only T1–T3 and T6–T8 have results from the ETEC306 phase. T4 (intruder detection), T5 (false alarms) and T9 (identity) have NOT been run yet (planned November 26 – December 1). Every threshold that depends on these tests is a preliminary value.

Each design item carries one of these status labels:

Label	Meaning
VERIFIED	Measured on the real prototype; the date and result are given
IMPLEMENTED	Runs in software v6 on the Pi, but has no formal test result yet
PRELIMINARY / UNTESTED	Design value or default setting; the test or calibration that confirms it is still pending (work package and date given)
PAPER STUDY	Studied on paper only this semester; not built

The detailed design is stored on GitHub: **https://github.com/Makabakamanihong/ETEC306/tree/main/docs/etec326-milestone1** (only this link is submitted). Note on naming: "Milestone 1" here is the course deliverable. The project plan uses its own milestone codes M0–M5 (Section 1.2).

1. Emerging Project Management Decisions

1.1 Resourcing

The project has one team member. The school has compressed ETEC306 and ETEC326 into one semester, so all remaining work is planned for one person with no over-allocation.

Resource	Role in the project	Decision
Tang Li (student)	Project manager, hardware, software, cloud, tests, documentation	Owns every work package; about 18.5 h per week from November 9 to December 11 (92.5 h planned); one work package per day
Instructor	Reviewer / approver	Reviews milestones and scope changes; [Instructor Name]
Lab technician (role)	Consulted	Only for equipment (USB power meter, bench supply) if the lab has it
Test volunteers (role)	Participants in T4 and T9	Non-enrolled people, only with written consent; no names are recorded
Hardware	Prototype from the ETEC306 phase	No new electronics; only a rigid mount and a simple cover (WP 2.2.1, 2.2.2)
Cloud	Oracle Cloud free-tier VM	Kept: $0 cost; Mosquitto, logger and WhatsApp forwarder already run there

Resource	Role in the project	Decision
Tools	GitHub, MS Project / Excel, Python, OpenCV, LibreOffice	All free or already available

1.2 Scheduling

The ETEC306 phase (concept, literature review, requirements, system design and prototype v1–v6) ran from September 21 to October 23, 2026. The remaining ETEC326 work runs from November 9 to December 11, 2026, with a two-day buffer to December 15. Each work package is 1–2 days, so the whole remaining chain is the critical path; the buffer is the only float.

ID	Milestone	Date	Status
M0	ETEC306 phase complete (working prototype v6)	Fri Oct 23, 2026	Achieved
M1	Revised project plan submitted	Mon Nov 9, 2026	Submitted
M2	Calibrated, hardened release v7 (feature freeze)	Fri Nov 20, 2026	Planned
M3	Verification complete (T1–T9)	Wed Dec 2, 2026	Planned
M4	Final report draft	Wed Dec 9, 2026	Planned
M5	Final demo and submission	Fri Dec 11, 2026	Planned

Design work that this report leaves open is scheduled as follows: wiring audit and power check (WP 2.1.1, Nov 10), rigid mount (2.2.1, Nov 11), radar–camera calibration (3.1.1, Nov 12–13), tracking tuning (3.1.2, Nov 16), enrolment and SFace threshold (3.1.3, Nov 17–18), simple cover (2.2.2, Nov 19), hardening and release v7 (3.2.1, Nov 20), tests T1–T9 (4.x, Nov 23 – Dec 1).

1.3 Scope decisions

- **Feature freeze on v6.** No new features after November 9. Release v7 (November 20) only adds calibration results, tuned settings and hardening. After v7 only fixes for failed tests are allowed.
- **Kept in scope:** radar–camera calibration, tuning, enrolment and threshold calibration, hardening, tests T1–T9, rigid mount, simple cover, manuals, final report and demo.
- **Moved to a paper study:** custom PCB, port to an RV1106 camera SoC (NPU), stepper-motor gimbal. These appear only in the cost-down memo (WP 5.1.1, Dec 3).
- **Dropped:** 3-D printed enclosure (replaced by a simple cover), new user features.
- **Cost basis:** Raspberry Pi 4B bought for $90 (currency assumed CAD), other parts about 180 CNY (estimate, about $34 CAD), Oracle Cloud free tier $0. The power bank is a temporary demo supply and is not counted.

1.4 Scope change log

All changes since the ETEC306 concept. Dates come from file timestamps, the software README and my project notes. As a solo project I approve my own changes; the instructor reviews them at each milestone.

ID	Date	Change	Reason	Impact
SC-01	Oct 5	Bench test page replaced by the formal program (etec306-app); bench service disabled, kept for rollback	Bench was only for wiring checks	One process owns all hardware
SC-02	Oct 5	Added vision-priority tracking, object recognition, calibration mode, patrol, HTTPS enrolment, settings page, autostart (v3)	Radar-only pointing missed faces; enrolment from a phone needs HTTPS	More CPU load; vision becomes main pointing source
SC-03	Oct 6	Pan range widened from ±45° to ±80° (611–2389 µs); side-face handling; phone notifications (v4)	Room coverage; real people turn their heads	Step test done; notifications off by default
SC-04	Oct 20	Identity gating fix, active search around the radar bearing, EN/ZH UI, ntfy / e-mail / Discord (v5)	Repeated false “unknown” alarms; camera often missed the radar target	Fewer false alarms (to be verified in T5/T9)
SC-05	Oct 20	CallMeBot dropped; self-hosted WhatsApp forwarder (Baileys) on the cloud VM	CallMeBot was full and could not add users	Unofficial API risk (R5); Telegram/ntfy as fallback
SC-06	Oct 20	Camera idle sleep, CPU and estimated power metrics, Wi-Fi set-up hotspot (v6)	Power, report data, demo in a new location	Power figures are estimates only
SC-07	Nov 9	Solo team confirmed; both courses compressed to finish by mid-December	School schedule	Re-plan; 92.5 h in 5 weeks
SC-08	Nov 9	Feature freeze; PCB / RV1106 / steppers to paper study; 3-D enclosure dropped	Time available to one person	Lower risk; product cost only estimated

1.5 Design risks carried into the detailed design

Risk	Design response
Under-voltage: the Pi reported an under-voltage flag since boot on Oct 20 (v6)	Separate servo supply (already), 5 V 3 A Pi supply, power check in WP 2.1.1, cable/port check
Servo reversal deadband ≈ 2° (measured Oct 6) is larger than the 0.44° PWM step	Report both values honestly (TR-10); vision closed loop with 25 px deadband; steppers only in the paper study
Identity threshold not yet calibrated for this camera and room	WP 3.1.3 with held-out photos; prefer a false “unknown” (extra alarm) over accepting a stranger
Unofficial WhatsApp library may stop working	Notification channel is a replaceable block; Telegram / ntfy / e-mail / Discord already in notify.py
T4/T9 need non-enrolled volunteers	Ask in week 1; consent form; report the real sample size

2. Enabling Technologies

For each technology: what it is, why I chose it, and how the design uses it. Status is given where it matters.

2.1 HLK-LD2450 24 GHz FMCW radar and its UART protocol

What	A 24 GHz FMCW (frequency-modulated continuous-wave) radar module with one transmit and two receive antennas. It tracks up to three moving people in the horizontal plane and reports X, Y (mm), radial speed (cm/s) and a distance-resolution value for each.
Why chosen	It sees people who stand almost still, works in the dark and through thin plastic, gives position (not just “motion” like a PIR sensor), and is cheap. Range is about 6 m and ±60° azimuth.
How used	Connected through a CH340 USB-UART adapter at 256000 baud, 8N1 (/dev/ttyUSB0). radar.py finds 30-byte frames: header AA FF 03 00, three 8-byte target slots, tail 55 CC. Each 16-bit value uses sign-magnitude (MSB set = positive). Empty slots are all zero. Targets with Y ≤ 0, range > 6 m or |azimuth| > 60° are rejected. The parser resynchronises when USB splits a frame. Measured ≈ 19 Hz frame rate (VERIFIED, ETEC306 phase). The module has no elevation output, so tilt comes from geometry (Section 4.4).

2.2 CH340 USB-UART bridge

What	A USB-to-serial converter (USB ID 1a86:7523).
Why chosen	Keeps the Pi's GPIO14/15 console UART free and reliably opens 256000 baud; some PL2303 clones cannot.
How used	Plugged into a Pi USB 2.0 port; powered by the Pi, not by the servo rail.

2.3 PCA9685 16-channel PWM driver (I2C)

What	An I2C chip that generates 16 hardware-timed PWM outputs with 12-bit resolution.
Why chosen	Linux software PWM jitters when the CPU is busy with vision; the PCA9685 keeps the 50 Hz servo pulses exact (TR-05).
How used	I2C-1 at address 0x40 (0x70 seen on the bus is its all-call address). Initialisation: MODE2 = 0x04, MODE1 = 0x30 (sleep + auto-increment), PRESCALE = 121 (50 Hz from 25 MHz), wait 5 ms, MODE1 = 0x20, wait 5 ms, MODE1 = 0xA0 (restart). After that only the four LEDn registers of CH4 (pan) and CH8 (tilt) are written. The chip is never put back to sleep and the ALL_LED registers are never used. 1 LSB = 20 ms/4096 ≈ 4.88 µs ≈ 0.44°.

2.4 MG90S micro servos

What	Metal-gear hobby servos, position set by a 1–2 ms pulse every 20 ms.
Why chosen	Small, cheap and strong enough for a USB camera; already part of the pan-tilt kit.
How used	Pan on CH4, tilt on CH8, powered from a separate 5 V supply. Pulse = 1500 − angle × 1000/90 µs. Measured on Oct 6: reversal deadband 20–29 µs (≈ 2°); 30° step settles in 254 ms median / 318 ms max including camera delay (VERIFIED).

2.5 USB UVC camera with MJPEG

What	A standard USB Video Class camera (identified as XC-CAMERA) that only offers MJPEG modes, from 320×240 up to 2960×1664, all at 30 fps.
Why chosen	Needs no driver, and MJPEG lets the Pi forward frames without decoding them.
How used	camera.py runs v4l2-ctl and splits the stream into JPEG frames (pass-through, 1280×720 at 15 fps). Detection decodes at half size (640×360). For identity the stream is paused and one still is taken at 2960×1664. The camera sleeps after 60 s with no target (stream closed; USB power stays on because the radar adapter shares the USB bus) and wakes as soon as the radar has a track.

2.6 OpenCV DNN: YuNet face detector and SFace face recogniser

What	Small ONNX models from the OpenCV Model Zoo, run with cv2.dnn on the Pi's CPU. YuNet finds faces and five landmarks; SFace turns an aligned face into a 128-number feature vector.
Why chosen	They run in real time on a Pi 4B without a GPU, and faces never leave the device (privacy, TR-33).
How used	YuNet runs at up to 8 Hz on 640×360 frames (face score ≥ 0.8). Its pixel error drives vision tracking. For identity, YuNet runs on the full-resolution still, the face is aligned (alignCrop) and SFace computes the feature. It is compared by cosine similarity with each enrolled person (Section 4.4 g). Enrolled features stay in faces/*.npz on the Pi.

2.7 Object and hand detection: NanoDet-Plus and MediaPipe palm detector

What	NanoDet-Plus-m (COCO, 80 classes) and the MediaPipe palm detector, both from the OpenCV Model Zoo.
Why chosen	The radar also reports chairs that move with a fan, curtains, etc. A visible body or hand confirms a person even when the face is turned away; a known object explains a non-human radar target.
How used	objects.py runs at about 1.5 Hz when vision is active. Person ≥ 0.55 confidence for 2 detections (3 without a matching radar track) confirms a person. A radar track that the camera looks at for 2.5 s (after the search pattern) with no person is marked “object” and ignored for 60 s. When the face turns away, a low-score face, the OpenCV profile-face Haar cascade and a KCF tracker keep following the head for up to 3 s.

2.8 Kalman filter (constant velocity)

What	A recursive estimator that combines a motion model with noisy measurements.
Why chosen	LD2450 points jump between frames (arms, legs, multipath). The filter smooths position and estimates velocity for a pointing lead.
How used	tracker.py runs one 1-D filter per axis with state [position, velocity]. Process noise comes from a walking acceleration of 1 m/s², measurement noise 100 mm. Innovations larger than 0.35 m are down-weighted unless they persist for 3 frames. The velocity gives a 0.25 s lead to compensate for gimbal lag.

2.9 Web server, MJPEG stream and HTTPS

What	Python’s standard ThreadingHTTPServer (no Flask), serving HTML pages, a JSON API and a multipart MJPEG video stream.
Why chosen	No extra framework to install on the Pi; one process holds all state. Browsers only allow camera access (for enrolment from a phone) over HTTPS.
How used	HTTP on port 8080 and HTTPS on 8443 (self-signed certificate) in the same process: dashboard, settings, enrolment, calibration, servo test and metrics pages; English by default with a Chinese switch. Secrets are never returned by the API. LAN only; there is no login yet (noted as a limitation).

2.10 MQTT over TLS, Mosquitto broker with ACL on Oracle Cloud

What	MQTT is a publish/subscribe protocol for small messages. Mosquitto is the broker; it runs on a free-tier Oracle Cloud VM.
Why chosen	Event-driven and light, with retained status, last-will (offline detection) and QoS 1. TLS protects the link; the ACL limits each account to its own topics.
How used	The Pi (paho-mqtt) connects to port 8883 with TLS 1.2, a username/password and the project CA. Anonymous access is off; port 1883 is not open. Topics and the ACL are in Section 4.5. A cloud logger (etec306-cloud) stores all messages as daily JSONL files for 30 days and computes daily statistics. QoS 1 events are queued while offline and sent after reconnect (T6 passed).

2.11 Phone notifications: Baileys WhatsApp forwarder, Telegram, ntfy, e-mail, Discord

What	Baileys is an open-source Node.js library that links to WhatsApp as a “linked device”. Telegram Bot API, ntfy, SMTP e-mail and Discord webhooks are alternatives.
Why chosen	The official WhatsApp Cloud API needs a Meta business account and template approval, and CallMeBot was full. A self-hosted forwarder works with a normal secondary WhatsApp account.

How used	etec306-wa runs on the cloud VM as an unprivileged user with no listening ports. It subscribes with a read-only MQTT account to the event and image topics and forwards alarm events with the photo, at most one per minute, to the configured recipient. It was linked by scanning a QR code on Oct 20. The Pi-side notify.py offers Telegram, ntfy, e-mail and Discord; all are off by default, and only one side should be enabled to avoid duplicates.

2.12 NetworkManager hotspot Wi-Fi provisioning

What	A set-up mode in which the Pi becomes a Wi-Fi access point with a small captive portal.
Why chosen	The device is headless. In a new place (for example the demo room) it must join Wi-Fi without a keyboard or screen.
How used	If no Wi-Fi or Ethernet connects within 60 s of boot, or K2 + K3 are held for 5 s, or the settings page asks for it, the Pi starts the hotspot “ETEC306-Setup-XXXX” (XXXX = last MAC digits) with a random 8-digit WPA2 password shown on the OLED. All DNS names point to 10.42.0.1. The user picks a network and types the password; nmcli saves it. A Wi-Fi QR code shown to the camera also works. After 10 minutes without success, the hotspot stops and the previous connection returns.

2.13 systemd services

What	The Linux service manager.
Why chosen	Automatic start at boot, restart after a crash, and a clean shutdown without a logged-in user.
How used	Pi: etec306-app (Restart=always, runs as the normal user with the gpio, i2c, video and dialout groups), etec306-wifi-check and etec306-wifi-setup. Cloud: mosquitto, etec306-cloud, etec306-wa. The old etec306-bench unit is disabled and conflicts with etec306-app.

2.14 GPIO with lgpio, I2C with smbus2, SSD1306 OLED

What	lgpio drives the Pi’s GPIO pins; smbus2 talks I2C; the SSD1306 is a 128×32 monochrome OLED controller.
Why chosen	lgpio works on current Raspberry Pi OS and supports pull-ups; the OLED shows status without a monitor.
How used	Four keys (internal pull-up, active low, 30 ms debounce), three LEDs (active high), OLED at I2C 0x3C sharing the bus with the PCA9685 behind one i2c_lock. The OLED has 4 pages (IP/state, MQTT, radar target, pan/tilt) and shows only ASCII.

2.15 Power estimation model

What	A linear model of whole-system power, because the Pi 4B has no power-measurement chip.
Why chosen	Gives the report and the dashboard a power figure until a USB meter can be used.
How used	metrics.py: P = 2.85 W idle + 3.5 W × CPU load + camera (0.85 W streaming / 0.25 W asleep) + radar 0.55 W + OLED/LEDs; the servos are reported separately. Values are logged every 10 s to a CSV. All coefficients are typical values or assumptions (ESTIMATE, not measured).

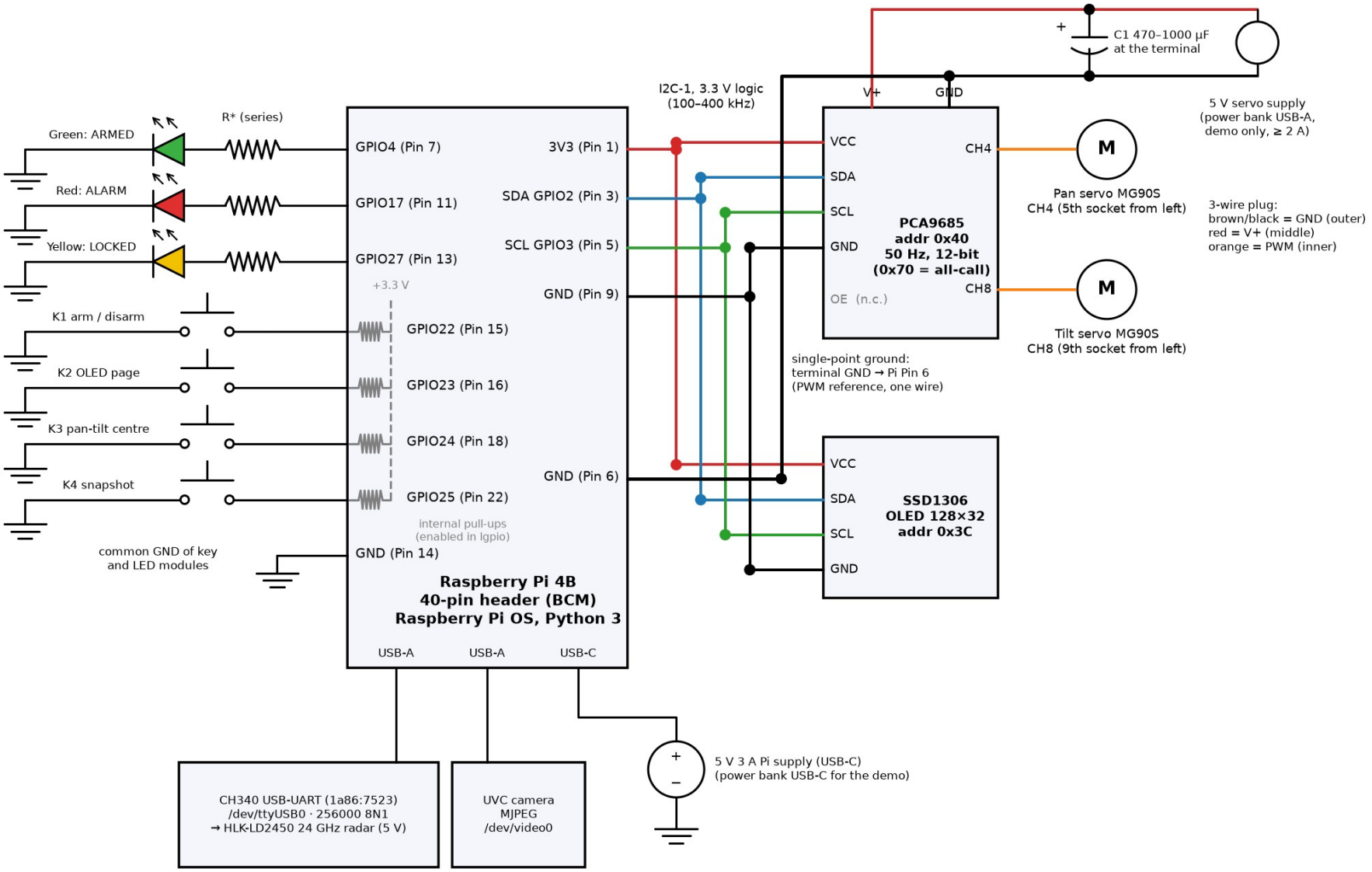
3. Preliminary Detailed Design – Hardware

Hardware status: the wiring below is the bench-verified wiring used by v6 (IMPLEMENTED; channel and pin checks done on the bench in the ETEC306 phase). A full wiring audit with a continuity check and a power check is WP 2.1.1 (Nov 10). The mechanical mount is PRELIMINARY (WP 2.2.1, Nov 11).

3.1 Pin map and interfaces

Signal / device	Pi pin	BCM	Type / level	Connected to	Status
3.3 V logic supply	1	–	Power out	PCA9685 VCC, OLED VCC	Implemented
I2C SDA	3	GPIO2	I2C-1, 3.3 V	PCA9685 SDA, OLED SDA	Implemented
I2C SCL	5	GPIO3	I2C-1, 3.3 V	PCA9685 SCL, OLED SCL	Implemented
GND (I2C devices)	9	–	Ground	PCA9685 GND, OLED GND	Implemented
GND (servo reference)	6	–	Ground, single wire	PCA9685 terminal GND	Implemented
Green LED – ARMED	7	GPIO4	Output, active high	Traffic-light module	Implemented
Red LED – ALARM	11	GPIO17	Output, active high	Traffic-light module	Implemented
Yellow LED – LOCKED (blinks in VERIFY)	13	GPIO27	Output, active high	Traffic-light module	Implemented
GND (keys, LEDs)	14	–	Ground	Common of key and LED modules	Implemented
K1 arm / disarm	15	GPIO22	Input, pull-up, low = pressed	Key module	Implemented
K2 OLED page	16	GPIO23	Input, pull-up, low = pressed	Key module	Implemented
K3 pan-tilt centre	18	GPIO24	Input, pull-up, low = pressed	Key module	Implemented
K4 snapshot	22	GPIO25	Input, pull-up, low = pressed	Key module	Implemented
K2 + K3 held 5 s	16 + 18	GPIO23/24	Key combination	Starts Wi-Fi set-up mode	Implemented
Radar	USB-A	–	UART 256000 8N1 via CH340	HLK-LD2450 (/dev/ttyUSB0)	Verified (≈ 19 Hz)
Camera	USB-A	–	USB UVC, MJPEG	Camera (/dev/video0)	Implemented
Pi power	USB-C	–	5 V, 3 A recommended	Supply (power bank for the demo)	Under-voltage seen – check 2.1.1

Device	Bus / address	Channel / setting	Notes
PCA9685	I2C-1, 0x40	CH4 = pan (5th socket from the left), CH8 = tilt (9th socket); 50 Hz	0x70 is the all-call address, not a third device; OE and the side V+ pin are not connected
SSD1306 OLED	I2C-1, 0x3C	128×32	Some modules use 0x3D; this one is 0x3C (i2cdetect)
Pan servo	PCA9685 CH4	611–2389 µs (±80°), centre 1500 µs + trim	Pulse up = left; step test Oct 6 (Verified)
Tilt servo	PCA9685 CH8	1000–2000 µs, soft limit ±25°	Pulse up = down
Servo supply	PCA9685 screw terminal	5 V, separate from the Pi; C1 470–1000 µF at the terminal	Single ground wire to Pi pin 6



Notes: GPIO14/15 (console UART) and GPIO2/3 are not used for keys/LEDs. Keys: pressed = LOW, 30 ms software debounce. LEDs: active HIGH, ≤ 8 mA per pin.
Radar, camera, OLED and PCA9685 logic are powered by the Pi; only the servos use the separate 5 V rail. Power up the Pi first, then the servo supply; remove servo power first.
Old draft pins (K1–K4 on GPIO17/27/22/23, LEDs on GPIO5/6/13, servos on CH0/CH1) are NOT used. * LED series resistor is on the traffic-light module; value [to confirm].

Figure 1 – Circuit schematic: Raspberry Pi GPIO header, PCA9685, servos with a separate 5 V supply and single-point ground, OLED, keys with pull-ups, LEDs with series resistors, radar via CH340 USB and USB camera

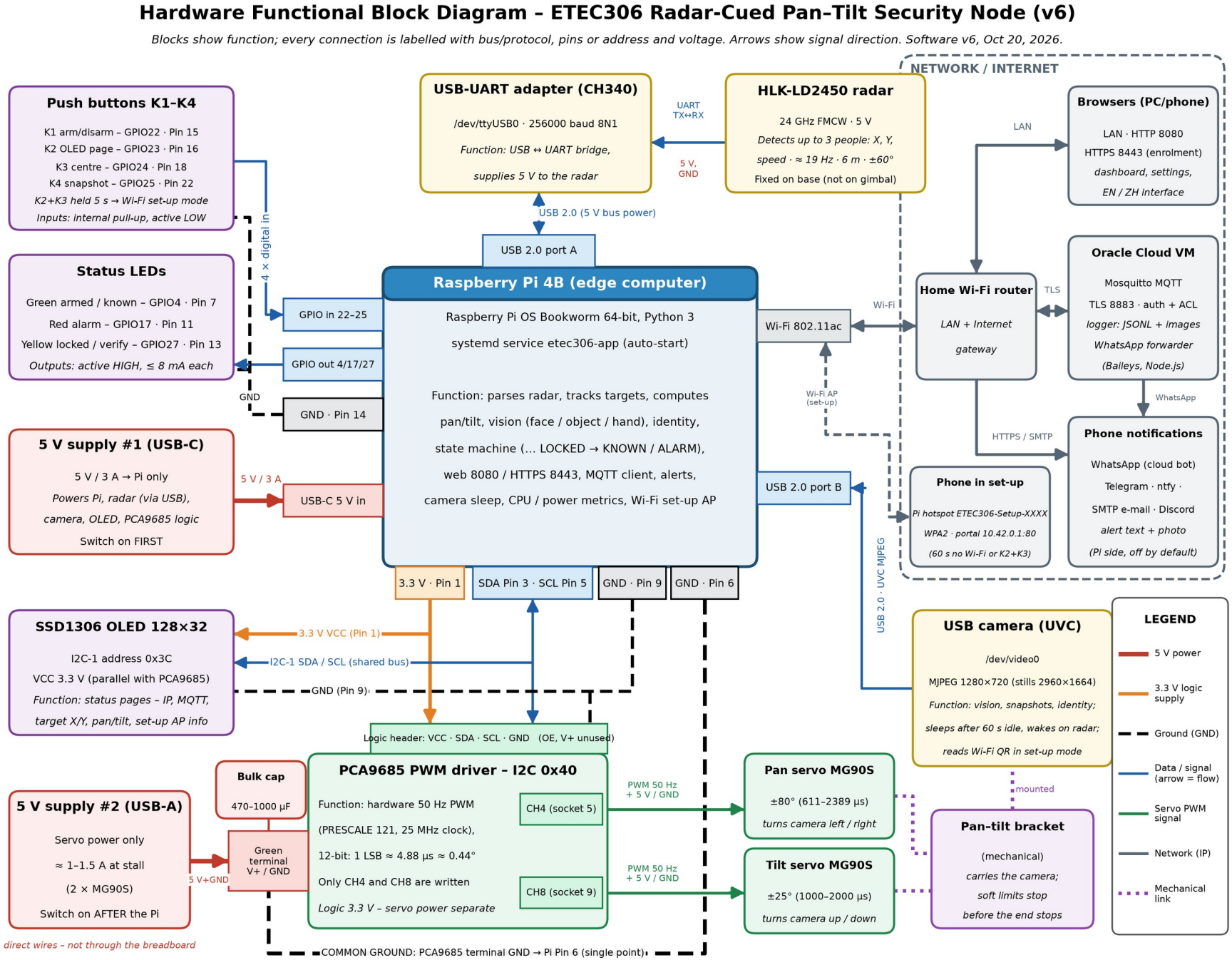


Figure 2 – Hardware functional block diagram (Assignment 5, updated to v6 in Assignment 5 v2)

3.2 Power design and budget

The logic side (Pi, radar, camera, OLED, PCA9685 logic) and the servo side use separate 5 V supplies, joined by one ground wire from the PCA9685 terminal to Pi pin 6. This keeps servo start-up current from resetting the Pi. The values below come from the linear model in metrics.py and are **ESTIMATES, not measurements**.

Load	Armed, idle (camera asleep)	Tracking (typical, 50 % CPU)	Worst case	Basis
Pi 4B board	2.85 W	4.60 W	6.35 W (100 % CPU)	Model: 2.85 W + 3.5 W × CPU
USB camera	0.25 W	0.85 W	0.85 W	Assumption (150–200 mA streaming)
LD2450 + CH340	0.55 W	0.55 W	0.55 W	Datasheet: supply ≥ 200 mA, typ. ≈ 100 mA
OLED + LEDs	0.08 W	0.10 W	0.12 W	Typical values
Logic total (5 V)	**3.7 W (0.75 A)**	**6.1 W (1.22 A)**	**7.9 W (1.57 A)**	Needs a 5 V 3 A supply
Servos (separate 5 V)	0.1 W	0.7 W (30 % moving)	≈ 5–7.5 W (both stalled, 1–1.5 A)	Assumption / design document

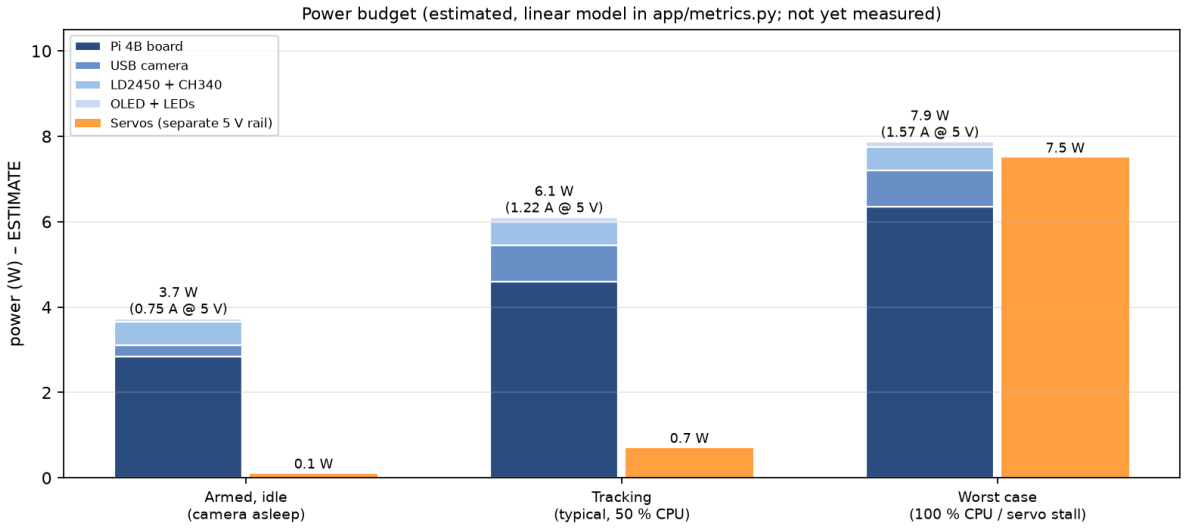


Figure 3 – Estimated power budget (not yet measured)

- **Pi supply:** 5 V, 3 A through USB-C. During the demo a power bank is used (temporary, not in the cost). With an energy E (Wh) the run time is about t ≈ 0.85 × E / P; the power-bank capacity is [to confirm].
- **Observation:** on October 20 the Pi reported an under-voltage flag since boot (vcgencmd get_throttled). During T7 (October 6) the flag stayed 0x0. WP 2.1.1 checks the cable, the port and, if a USB meter is available, measures real idle and tracking power to replace the estimates.
- **Servo supply:** two MG90S stall at about 1–1.5 A together; a 2 A or larger 5 V port is enough. C1 (470–1000 µF) at the terminal absorbs reversal current. Power-up order: Pi first, then the servo supply; power-down: servo supply first.

3.3 Mechanical mounting (PRELIMINARY)

- Radar and pan-tilt on **one rigid base** (WP 2.2.1, Nov 11). Foam may damp vibration but must not set the radar angle, because it creeps and the calibration drifts.
- Mounting height 1.2–1.5 m for the room demo (the current software default assumes 1.0 m; the real height is measured in WP 3.1.1). No large metal surfaces within 6 m in front of the radar (multipath makes coordinates jump).
- At pan = 0 the camera axis should point along the radar boresight. The remaining offsets are calibrated as θ_0 and φ_0 (v4 fit: $\theta_0 = -3.9°$, to be re-measured).
- Soft limits pan ±80°, tilt ±25° stop the gimbal before its mechanical end stops (stalling heats the servo).
- Simple cover (WP 2.2.2, Nov 19): plastic or card only; nothing metallic or metal-painted in front of the radar; openings for the camera, OLED, keys and LEDs; cable strain relief.

3.4 Hardware design notes

- Old draft pins (keys on GPIO17/27/22/23, LEDs on GPIO5/6/13, servos on CH0/CH1) must not be used; the table in 3.1 is the bench-checked wiring.
- Sockets are counted from the left: the 5th socket is CH4 and the 9th is CH8. Counting from the right gives CH11 and CH7 (an earlier mistake).
- Clone PCA9685 boards have a few per cent oscillator error, so the real centre is trimmed with pan/tilt offset (µs) during calibration instead of trusting 1500 µs.
- If both I2C boards have pull-ups, the bus pull-up is stronger; 400 kHz still works, otherwise drop to 100 kHz.
- GPIO14/15 (console) and GPIO2/3 (I2C) are not used for keys or LEDs. Each LED pin stays below 8 mA.

4. Preliminary Detailed Design – Software

Software status: v6 is IMPLEMENTED and runs as a systemd service on the Pi. All thresholds in this section are the current defaults; they become final only after calibration and tuning (WP 3.1.1–3.1.3, Nov 12–18) and tests T4, T5 and T9.

4.1 Module architecture

Module (app/)	Responsibility	Thread / rate
app.py	Main program: state machine, guidance, identity loop, HMI, MQTT loop, web server, logging	Several threads (below)
radar.py	LD2450 frame parser and serial receive thread	Every frame (≈ 19 Hz)
tracker.py	Point merging, nearest-neighbour association, Kalman filter, primary-target choice	Per radar frame
pointing.py	Radar X/Y → pan/tilt angles → pulse widths; limits; pixel → angle	Pure functions
hw.py	PCA9685 and SSD1306 drivers (shared i2c_lock), keys and LEDs (lgpio)	Guidance 50 Hz, HMI 5 Hz, keys 100 Hz polling
camera.py	MJPEG pass-through, full-resolution stills, sleep / wake	Camera stream 15 fps
vision.py	YuNet face detection, head tracking, SFace gallery and matching, enrolment	≤ 8 Hz

Module (app/)	Responsibility	Thread / rate
objects.py	NanoDet-Plus objects and palm detector	≈ 1.5 Hz
settings.py	Live-tunable parameters with limits, saved to settings.json	On change
mqtt_client.py	TLS connection, heartbeat, track, event, image, commands, offline queue	Heartbeat 10 s, track ≤ 2 Hz
notify.py	Telegram / ntfy / e-mail / Discord (off by default); secrets in notify.json (chmod 600)	Per alarm, ≤ 1/min
metrics.py	CPU, temperature, throttling flags, estimated power; CSV every 10 s	10 s
tools/wifi_setup.py	Hotspot, captive portal, nmcli connect (separate root service)	On demand
server/ (cloud)	Mosquitto config and ACL, etec306-cloud logger, statistics, command tool, WhatsApp forwarder	systemd services

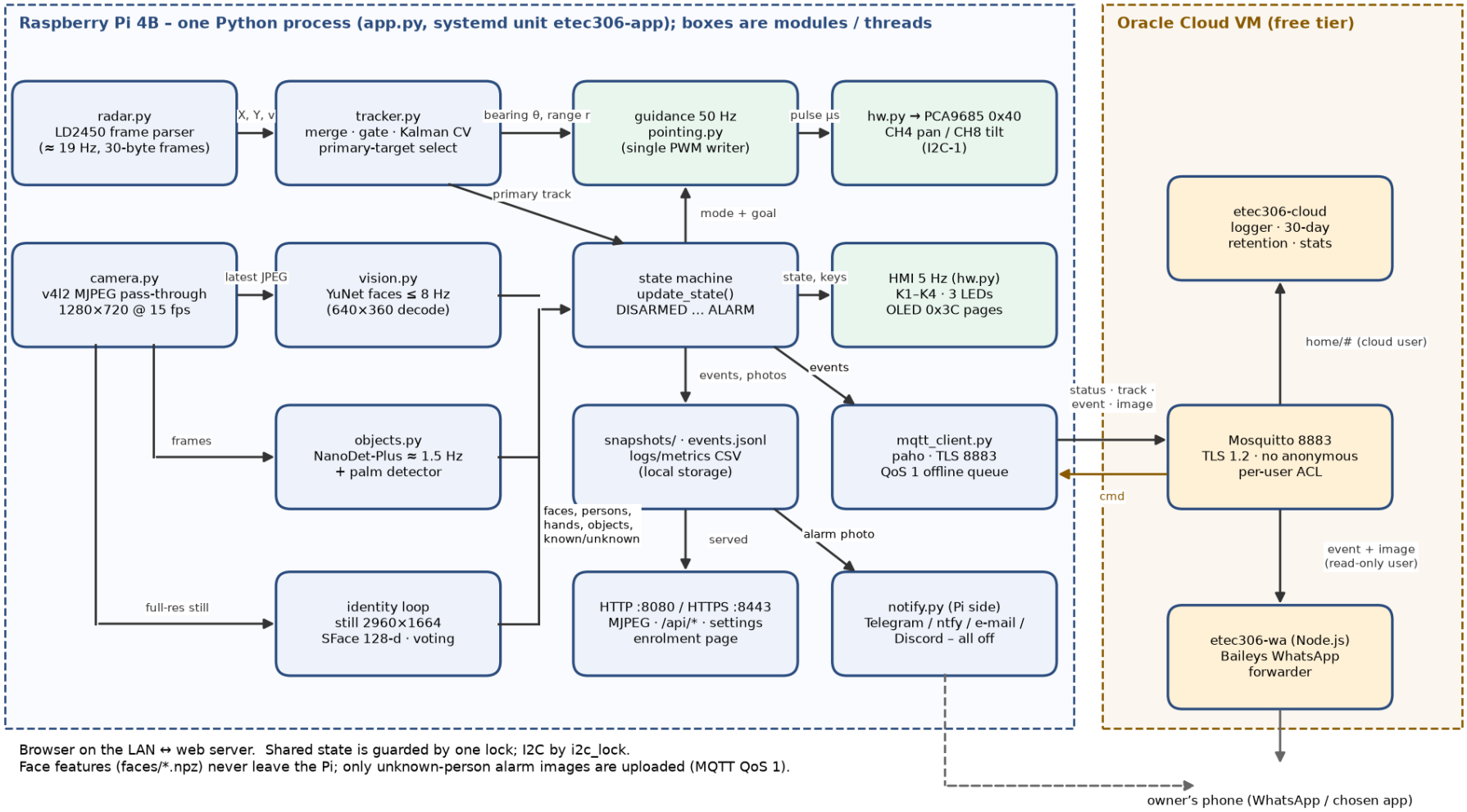


Figure 4 – Software data flow between modules on the Pi and the cloud

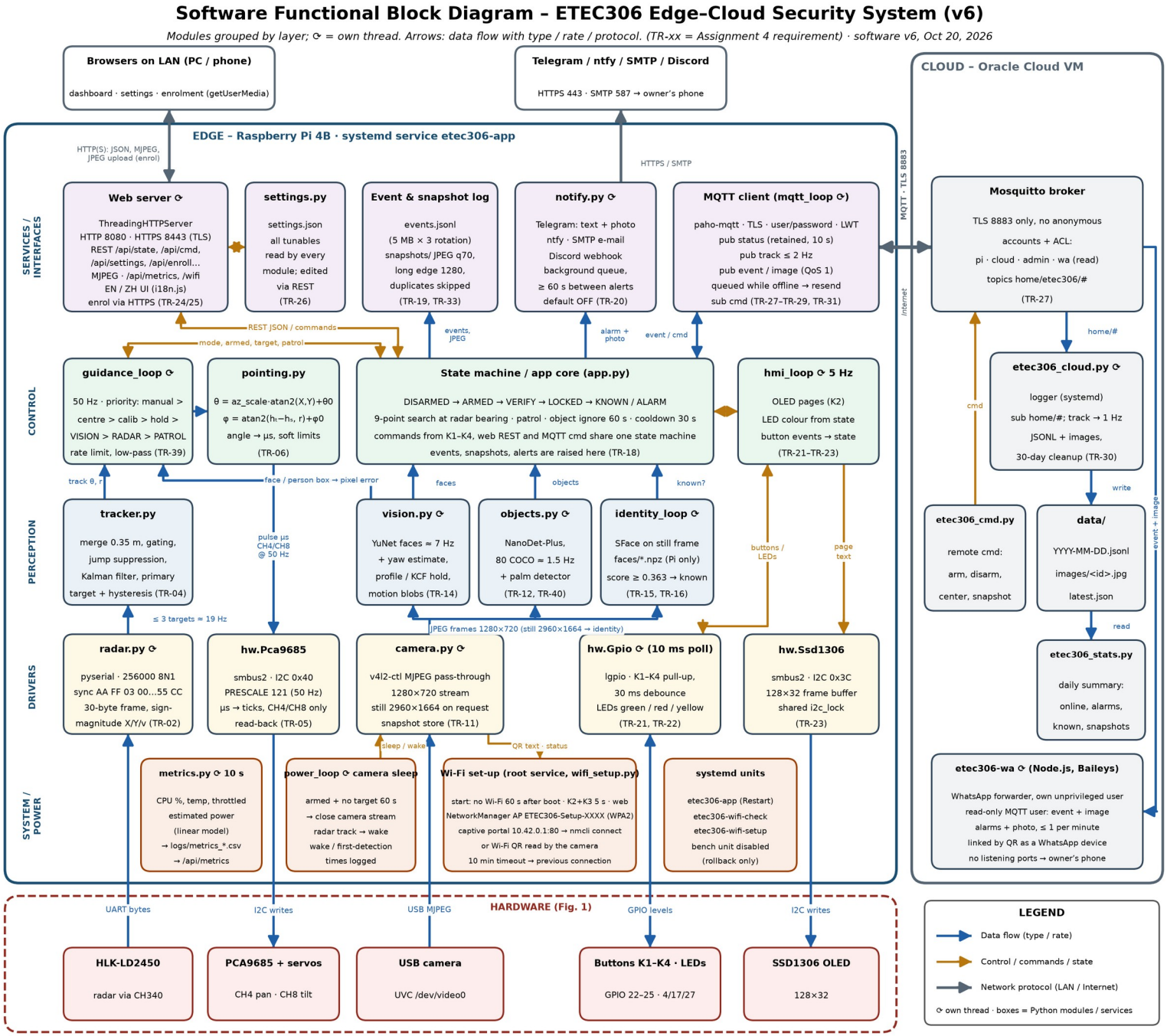


Figure 5 – Software functional block diagram (Assignment 5, updated to v6 in Assignment 5 v2)

4.2 System state machine

The state machine in update_state() runs inside the 50 Hz guidance loop. The design document used the names IDLE / SEARCH / TRACK; in the code these behaviours are split into the system state (below) and the guidance mode (Section 4.3: patrol, search, radar, vision). Searching and tracking are therefore guidance modes inside ARMED, VERIFY and LOCKED.

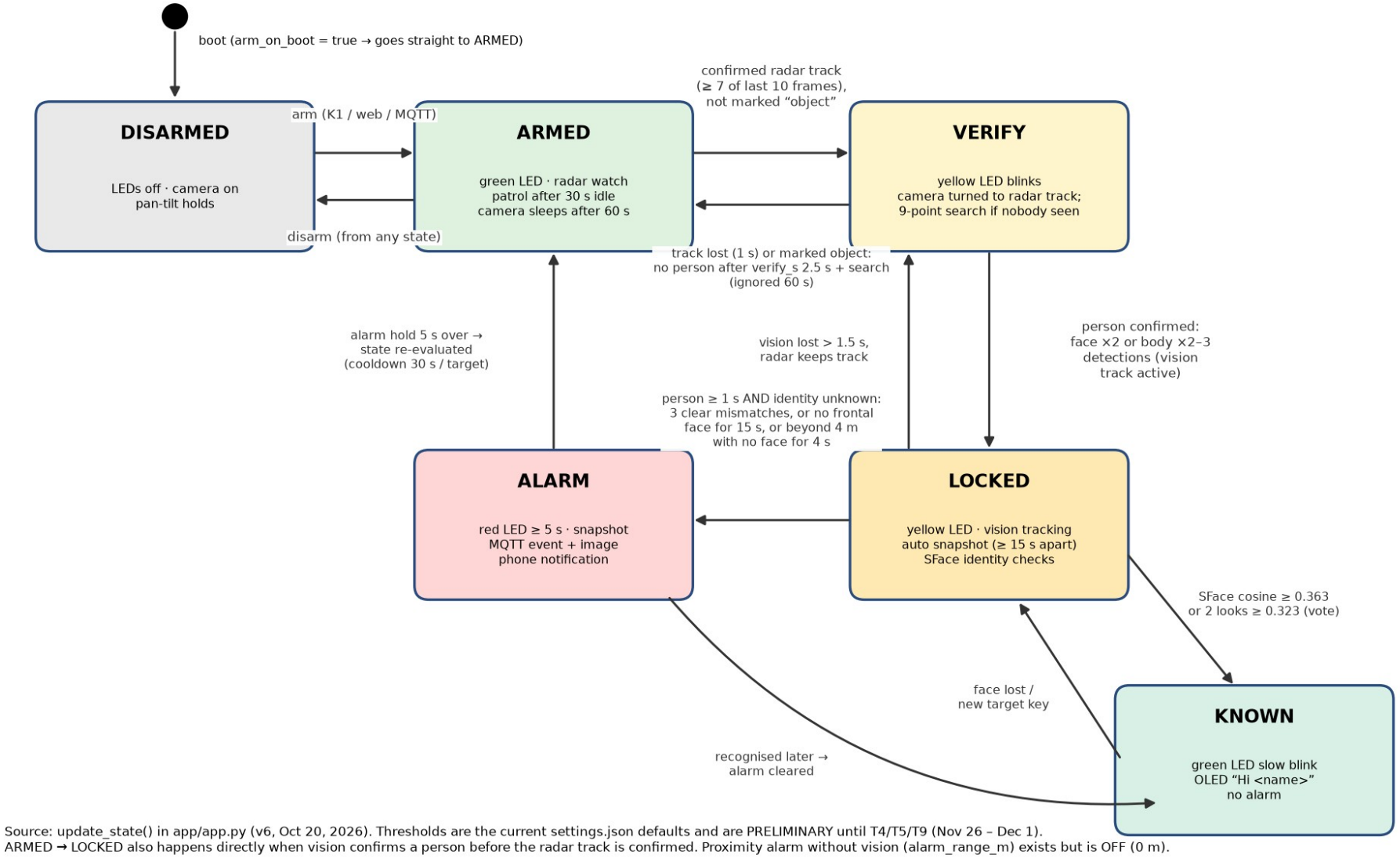


Figure 6 – System state machine (from app.py, v6)

State	Entered when	Outputs
DISARMED	Disarmed by K1, web or MQTT command	LEDs off; pan-tilt holds; no alarms
ARMED	Armed, no confirmed non-object radar track and no person	Green LED; patrol after 30 s idle; camera sleeps after 60 s
VERIFY	Confirmed radar track (≥ 7 of last 10 frames) that is not marked “object”	Yellow LED blinks; camera turned to the track; search pattern if nobody is seen
LOCKED	Vision confirms a person (face ×2 or body ×2–3)	Yellow LED; vision tracking; one auto snapshot (≥ 15 s apart); identity checks
KNOWN	LOCKED and the person matched an enrolled face	Green LED slow blink; OLED “Hi <name>”; no alarm
ALARM	Person for ≥ 1 s and identity unknown (see 4.4 g); held ≥ 5 s	Red LED; snapshot; MQTT event + image; phone notification; 30 s cooldown per target

4.3 Guidance arbitration

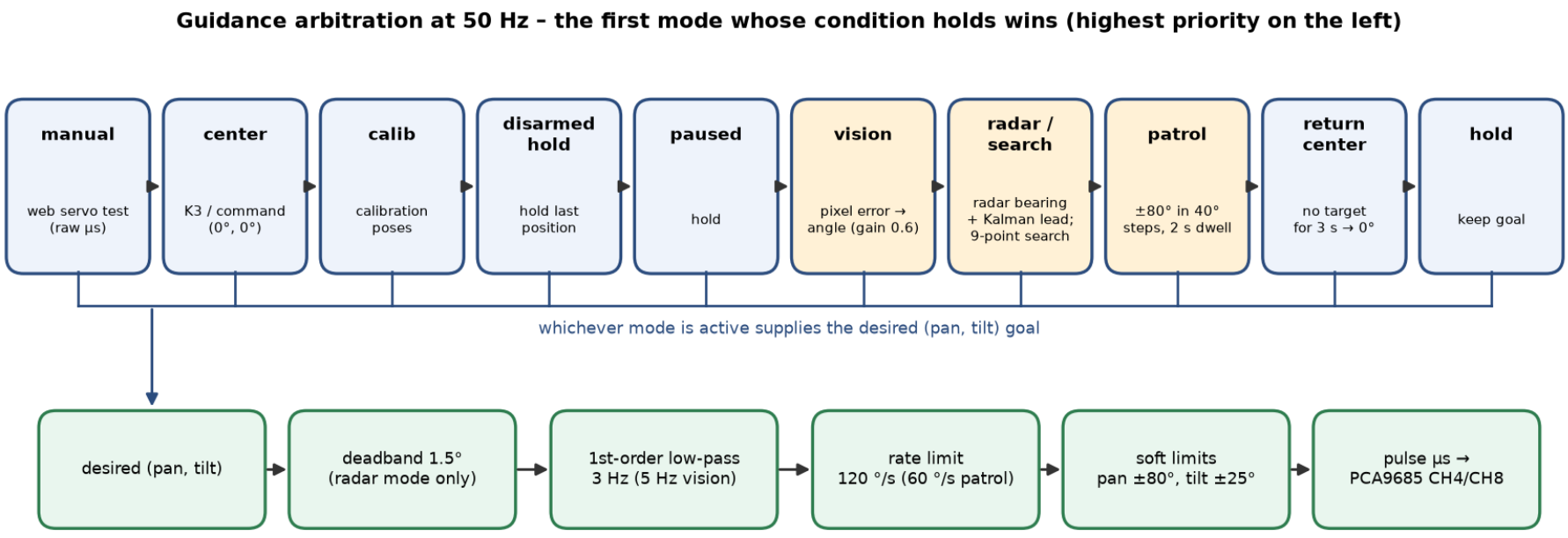


Figure 7 – Guidance modes in priority order and the command chain to the PCA9685

4.4 Key algorithms

a) Radar tracking. Points closer than 0.35 m are merged (arm and body of one person). Nearest-neighbour association with a 0.7 m gate; a track is confirmed when it appears in ≥ 7 of the last 10 frames and is dropped after 1 s without a measurement. Valid range 0.5–6 m. The primary target prefers vision-confirmed and approaching tracks, then the nearest; switching needs a 0.5 m advantage held for 0.8 s.

b) Radar → pan/tilt geometry. With s = x_sign and the mounting offsets θ_0, φ_0:

$$\theta = az_scale \cdot atan2(s \cdot X, Y) + \theta_0 \qquad \varphi = atan2(h_t - h_s, r) + \varphi_0, \quad r = \sqrt{X^2 + Y^2}$$

The LD2450 gives no elevation, so tilt is aimed at a target height h_t (default 1.3 m) from the sensor height h_s (default 1.0 m). Pan uses the Kalman position plus a 0.25 s velocity lead.

c) Pulse formula. For each servo:

$$u\ (\mu s) = clamp(1500 - sign \cdot angle \cdot 1000/90 + trim,\ u_min, u_max); \quad ticks = round(u \cdot 4096 / 20000)$$

Pan is clamped to 611–2389 µs (±80°), tilt to 1000–2000 µs with a ±25° soft limit. Right and up are positive. Example: 18° right = 1300 µs.

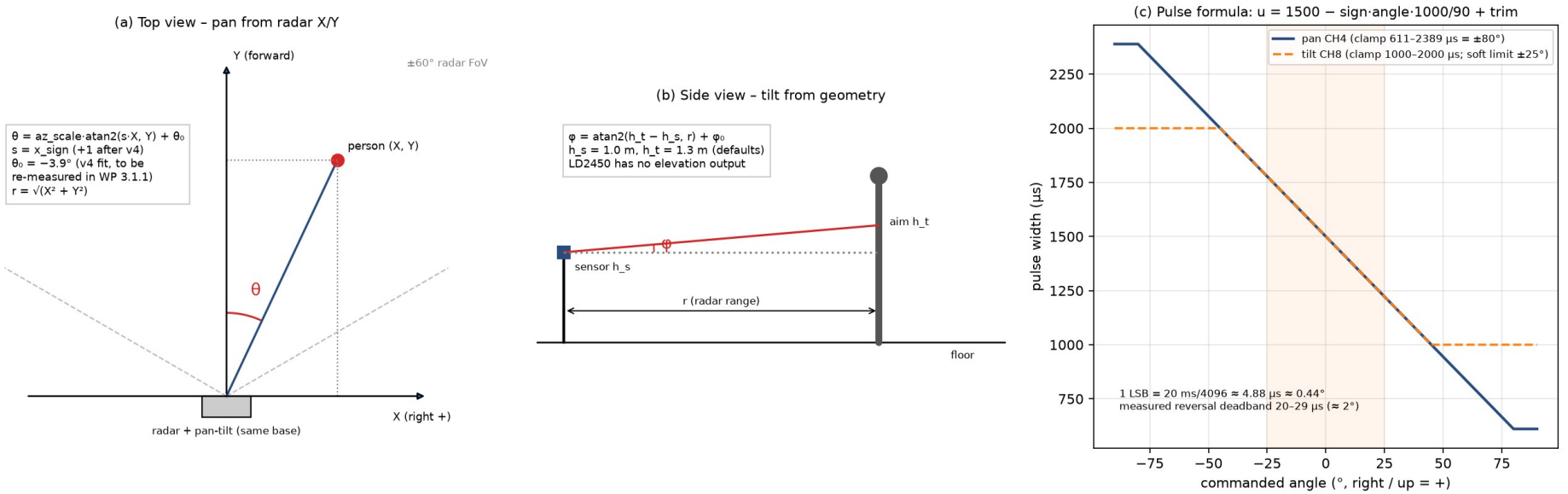


Figure 8 – (a) pan from radar X/Y, (b) tilt from geometry, (c) pulse-width mapping

d) Sign and offset calibration. (1) Direction self-test, no person needed: the gimbal moves 8° right and 8° up, and phase correlation of two frames shows which way the image moved. This sets pan_sign and tilt_sign (v3 result: +1/+1). (2) Point calibration: a person or a waved object is seen by both the radar and the camera at several positions. The software fits x_sign, az_scale and θ_0 (only the bias if the positions span less than 8°) and compares x_sign = ±1. The v4 fit gave x_sign = +1, θ_0 = −3.9°, az_scale = 1.0. **PRELIMINARY:** re-done in the final mount (WP 3.1.1, Nov 12–13).

e) Vision tracking. When a face or body is confirmed, the goal becomes $pan_0 + k \cdot \Delta\theta$, $tilt_0 - k \cdot \Delta\varphi$, where $pan_0/tilt_0$ is the commanded pose when the frame was taken, $\Delta\theta/\Delta\varphi$ is the pixel error converted with a pinhole model (HFOV 100°, VFOV 68° defaults), k = 0.6, with a 25 px deadband. Radar is ignored until vision is lost for 1.5 s.

f) Search and patrol. If the camera points at a radar track but sees nobody for 1.5 s, it steps through 9 offsets around the radar bearing: (0,0), (−d/2,0), (d/2,0), (−d,0), (d,0), (−d/2,t), (d/2,t), (0,t), (0,−t/2) with d = 18° and t = 10°, stopping 0.8 s at each. While still, frame differencing shows moving blobs to aim at. Only after a full pattern with no person is the track marked "object". With no target for 30 s, patrol sweeps ±80° in 40° steps with a 2 s stop (overlap with a ≈ 76° camera view).

g) Face matching and voting. Only near-frontal faces (|yaw| ≤ 60°, eye distance ≥ 40 px on the still, sharp enough: Laplacian variance ≥ 120) are compared. Score = the larger of cosine(feature, person mean) and the mean of the 3 closest samples; turned faces are also compared with the two best enrolled samples of the same side within 20° of yaw. Known if score ≥ 0.363 (+0.05 when |yaw| > 30°). Voting: two of the last four scores ≥ 0.323 for the same best person also count as known. Unknown after 3 clear mismatches (eye distance ≥ 60 px, |yaw| ≤ 30°), or when a close person shows no comparable face for 15 s. This "soft unknown" keeps comparing and is cleared by a later match. Beyond 4 m, a person with no face for 4 s raises the "unknown beyond identify range" alarm. For 120 s after a family member was recognised, a body without a face does not raise an alarm. **PRELIMINARY:** 0.363 is the OpenCV starting value; WP 3.1.3 calibrates it for this camera and room, and T9 tests it.

h) Power saving. In ARMED with no radar track or vision target for 60 s the camera stream is closed; any radar track wakes it. Wake-to-first-frame and wake-to-first-detection times are logged for the report.

4.5 Interfaces and APIs

MQTT topics (broker on the cloud VM, TLS port 8883; device id etec306):

Topic	Direction	QoS / retain	Payload
home/etec306/status	Pi → cloud	1, retained; last-will {"online": false}	Heartbeat every 10 s: armed, state, mode, pan/tilt, uptime, CPU temperature, radar Hz
home/etec306/track	Pi → cloud	0	Primary target X, Y, speed, bearing, source (radar/vision); ≤ 2 Hz

Topic	Direction	QoS / retain	Payload
home/etec306/event	Pi → cloud	1, queued offline	alarm / known / snapshot / arm events (metadata only)
home/etec306/image/<event_id>	Pi → cloud	1, queued offline	JPEG (quality 70, long edge 1280) for unknown-person alarms only
home/etec306/cmd	cloud → Pi	1	{"action": arm | disarm | toggle_arm | center | snapshot | pause | resume | clear_alarm}
home/etec306/wa/#	forwarder → Pi	1, retained	WhatsApp forwarder link status / link QR. The ACL file in the repository does not yet grant this write – to be checked in hardening (WP 3.2.1)

MQTT account	Allowed
etec306-pi (device)	write status, track, event, image/#; read cmd
etec306-cloud (logger, tools)	read home/#; write cmd
etec306-admin (debugging)	read/write home/#
etec306-wa (WhatsApp forwarder)	read home/etec306/event and home/etec306/image/# (read-only)

HTTP API (Pi, ports 8080 / 8443):

Endpoint	Method	Purpose
/api/state	GET	Live state, targets, identity, guidance mode (dashboard polling)
/api/stream.mjpg, /api/frame.jpg	GET	Live MJPEG stream / single frame (placeholder image while the camera sleeps)
/api/settings, /api/settings/reset	GET / POST	Read or change tunable parameters (validated against limits)
/api/cmd	POST	arm, disarm, centre, snapshot, pause, resume, clear alarm, identify, camera wake/sleep, Wi-Fi set-up, reset identity
/api/servo, /api/calib	POST	Manual servo test (raw µs), calibration actions
/api/enroll, /api/enroll/upload, /api/enroll/finish, /api/faces/delete	POST	Face enrolment from the Pi camera or a phone/PC camera (HTTPS)
/api/notify, /api/notify/test	GET / POST	Notification settings (secrets never returned), one test message
/api/metrics, /api/wifi, /api/wa_status	GET	CPU / power estimate, Wi-Fi state, WhatsApp forwarder state

Interface	Specification
Radar	USB-UART (CH340) 256000 8N1, 30-byte frames AA FF 03 00 … 55 CC
Servos	PCA9685 I2C 0x40, CH4/CH8, 50 Hz, 12-bit
OLED	I2C 0x3C, 128×32, 4 pages via K2
Camera	USB UVC, MJPEG 1280×720 at 15 fps stream; 2960×1664 stills
Local files	settings.json (parameters), mqtt.json (broker credentials), notify.json (channel secrets, chmod 600), faces/*.npz (features), snapshots/, events.jsonl, logs/metrics_YYYYMMDD.csv. Credentials are never committed to Git.

4.6 Configuration parameters (current defaults)

Group	Parameter = default	Status
Geometry	x_sign = +1 (v4 fit), pan_sign = +1, tilt_sign = +1, az_scale = 1.0, θ_0 = −3.9° (v4), φ_0 = 0, h_s = 1.0 m, h_t = 1.3 m	Preliminary – WP 3.1.1
Radar tracking	Kalman on, accel 1 m/s², meas 100 mm, merge 0.35 m, jump 0.35 m, gate 0.7 m, drop 1 s, min range 0.5 m, max 6 m	Preliminary – WP 3.1.2
Guidance	low-pass 3 Hz, max rate 120 °/s, radar deadband 1.5°, return to centre 3 s, lead 0.25 s, 50 Hz loop	Preliminary – WP 3.1.2, T3
Vision	YuNet ≤ 8 Hz, score 0.8; NanoDet 1.5 Hz, conf 0.4; person 0.55 × 2; hand 0.8; vision gain 0.6, deadband 25 px, lost 1.5 s; verify 2.5 s; object ignore 60 s	Preliminary – T5, T8
Identity	range 4 m, threshold 0.363, side margin 0.05, vote margin 0.04, min eye 40 px, max yaw 60°, 3 attempts, wait 15 s	Preliminary – WP 3.1.3, T9
Search / patrol	search ±18° / 10°, dwell 0.8 s, after 1.5 s; patrol after 30 s, ±80°, 40° steps, 2 s	Implemented

Group	Parameter = default	Status
Alarm	confirm 1 s, far delay 4 s, known hold 120 s, cooldown 30 s per target, hold 5 s, snapshot ≥ 15 s apart	Preliminary – T4, T5
Power / misc.	camera sleep 60 s, metrics every 10 s, stream ≤ 10 fps, MQTT track 2 Hz, arm on boot	Implemented

4.7 Software design notes

- **Single writer:** only the guidance thread writes CH4/CH8. Vision, calibration and web commands change goals, never PWM. It checks MODE1 every 2 s and re-initialises the PCA9685 if it went to sleep (for example after a logic-power glitch).
- **Locks:** one application lock guards shared state; i2c_lock serialises the PCA9685 and the OLED on the same bus.
- **Privacy and security:** face features stay on the Pi; only unknown-person alarm images are uploaded. MQTT uses TLS and per-user ACLs. Secrets live in files with mode 600 and are scrubbed from error messages. The web pages have no login yet (LAN only); this is a known limitation for the final report.
- **Robustness:** systemd restarts the app; MQTT reconnects with back-off (1–60 s) and flushes the QoS 1 queue (T6). The Wi-Fi set-up mode always returns to the previous connection after 10 minutes.
- **Rollback:** deployments keep the previous version as a backup on the Pi (v1, v2 and v3 backups are documented in the README).

5. Verification Status – What Is Still Preliminary

Test	What	Result so far	Status	Planned
T1	Radar frame rate and range	≈ 19 Hz (≥ 10 Hz met); range error not yet reported	Partly verified	Nov 24
T2	Smallest servo step	PWM 0.44°/LSB; reversal deadband 20–29 µs (≈ 2°), Oct 6	Verified (re-run on v7)	Nov 24
T3	30° step latency	Software 30 ms median / 50 ms max; to a settled image 254 ms median / 318 ms max incl. ≈ 100 ms camera delay, Oct 6	Verified (re-run on v7)	Nov 24
T4	Intruder true positives (≥ 30)	–	PRELIMINARY / UNTESTED	Nov 26–27
T5	False alarms (fan, curtain, empty room)	–	PRELIMINARY / UNTESTED	Nov 30
T6	Wi-Fi outage	Queued events sent after reconnect	Verified	Nov 25
T7	Servo stress, 2 min	110 sweeps, throttled 0x0, no reboot, register read-back within 2 µs, Oct 6	Verified	Nov 25
T8	Visual centring	Met (qualitative)	Partly verified	Nov 25
T9	Identity: 2 known + 2 unknown people	–	PRELIMINARY / UNTESTED	Dec 1

Also preliminary: the mechanical mount, the calibrated geometry (θ_0, φ_0, h_s), the identity threshold, the power figures (estimates), and the cover. New in v6 and not yet exercised in a formal test: Wi-Fi set-up mode, camera sleep/wake times, WhatsApp forwarding of an alarm photo.

6. Discarded Designs, Features and Aspects

Each item was really used or planned at some point and then replaced. Evidence: software README version notes, configuration comments, design document and my project notes.

#	Discarded	Replaced by	When	Rationale
D1	Draft pin map: keys on GPIO17/27/22/23, LEDs on GPIO5/6/13, servos on PCA9685 CH0/CH1	Bench-verified map (keys GPIO22–25, LEDs GPIO4/17/27, servos CH4/CH8)	Before v1 (bench)	The draft did not match the modules as wired; channels were found by moving each output on the bench
D2	Pan limit 1000–2000 µs (±45°); design-document soft limit ±70°	±80° (611–2389 µs)	v4, Oct 6	Wider room coverage; a step-out test showed the image still moved at both ends with no stall and read-back matched
D3	Tilt direction guessed as −1 from observation	Automatic direction self-test (image phase correlation)	v2 → v3, Oct 5	The guess was wrong and made the vision correction positive feedback (camera ran away)
D4	x_sign = −1 for the radar	x_sign = +1 with $\theta_0 = -3.9°$ (point fit)	v4, Oct 6	A person on the far left appeared on the right; measured bearings showed the sign was reversed
D5	Minimum-range mask of 0.6 m (static target treated as gimbal reflection)	min_range 0.5 m	v4, Oct 6	The static target at ≈ 0.6 m was actually me sitting at the desk; masking it hid a real person
D6	Permanent “unknown” after a 15 s timeout (no more comparisons)	Soft unknown that keeps comparing; multi-look voting; yaw-aware matching	v5, Oct 20	A seated person facing sideways became “unknown (0.00)” and caused an alarm every 30 s
D7	Radar-only pointing	Vision-priority closed loop; radar to (re)acquire; 9-point search	v3 / v5	Radar bearing error and gimbal lag left faces at the image edge; the camera often missed the target
D8	Bench test page (etec306-bench) as the running program	Formal program etec306-app	Oct 5	Bench read the radar in 0.3 s chunks (≈ 3 frames/s) and had no state machine; kept disabled for rollback
D9	CallMeBot WhatsApp API	Self-hosted Baileys forwarder on the cloud VM	Oct 20	The CallMeBot service was full and could not add my number; text only
D10	Official Meta WhatsApp Cloud API	Baileys forwarder; Telegram / ntfy / e-mail / Discord as alternatives	Oct 20	Needs a Meta business account, a business phone number and template approval – too much for a course demo
D11	Linking WhatsApp with a phone-number pairing code	QR-code linking	Oct 20	A pairing-code script was prepared, but the link was completed by scanning the QR code, which needs no phone number in the config
D12	Notifications sent only from the cloud (ntfy/Telegram in the cloud logger)	Pi-side notify.py (v4) and cloud WhatsApp forwarder; only one side enabled	v4–v6	The alarm photo is already on the Pi; a test button gives an immediate result; avoids duplicate messages
D13	MobileNet-SSD person detector (Haar kept only for profile faces)	NanoDet-Plus + palm detector	v3	80 classes (to name non-human objects such as a chair or laptop); the SSD model is no longer loaded
D14	Real-time inference on full-resolution YUYV video	720p MJPEG pass-through; 640×360 detection; one 2960×1664 still for identity	Design / v5	YUYV at 2K is far too slow on a Pi 4B; stills give the resolution where it matters
D15	60 Hz power-line setting on the camera	Camera defaults	v5, Oct 20	It switched the camera to 30 fps and raised CPU load
D16	Custom PCB	Breadboard + modules; PCB only in the paper cost-down study	Nov 9	Not possible for one person in 5 weeks with tests still to do
D17	RV1106 camera SoC port	Raspberry Pi 4B; RV1106 in the paper study	Nov 9	Porting the vision stack to an NPU is a project of its own
D18	Stepper-motor gimbal	MG90S servos; steppers in the paper study	Nov 9	Would remove the ≈ 2° deadband but needs new drivers and mechanics
D19	3-D printed enclosure	Simple plastic/card cover	Nov 9	Print time and iterations; the radar needs a non-metal window anyway

7. Plan to Milestone 2

- Post the datasheets in Appendix A as PDFs in the GitHub folder and attach them as an appendix to the Milestone 2 report.

- Replace preliminary values with calibrated ones (θ_0, φ_0, h_s, SFace threshold) and record them in the change log.
- Release v7 on November 20 and run T1–T9 against it; update Section 5 with measured results, including power if a USB meter is available.

Appendix A – Datasheets to Be Posted in Milestone 2

Component	Document	Source (link)
HLK-LD2450 radar	Product page / datasheet; serial protocol manual V1.03 and parse demo (already in the project docs/ folder)	https://www.hlktech.net/index.php?id=1157
PCA9685	NXP PCA9685 16-channel 12-bit PWM controller datasheet	https://cdn-shop.adafruit.com/datasheets/PCA9685.pdf
SSD1306	Solomon Systech SSD1306 OLED controller datasheet	https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf
Raspberry Pi 4B	Product brief and datasheet (GPIO, power)	https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-product-brief.pdf ; https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-datasheet.pdf
MG90S servo	TowerPro MG90S specification	https://www.towerpro.com.tw/product/mg90s-3/ ; https://www.electronicoscaldas.com/datasheet/MG90S_Tower-Pro.pdf
CH340 USB-UART	WCH CH340 datasheet	https://www.wch-ic.com/products/CH340.html ; https://cdn.sparkfun.com/datasheets/Dev/Arduino/Other/CH340DS1.PDF
USB camera	UVC camera (XC-CAMERA) specification	[to locate – seller listing]
Models / software	YuNet, SFace, NanoDet-Plus, palm detector (OpenCV Zoo); Mosquitto configuration manual; Baileys	https://github.com/opencv/opencv_zoo ; https://mosquitto.org/man/mosquitto-conf-5.html ; https://github.com/WhiskeySockets/Baileys

All links were checked on November 9, 2026. The NXP website blocked the automatic check, so the PCA9685 datasheet link points to a copy hosted by Adafruit.

Appendix B – Sources

- My ETEC306-phase documents: Idea of the project; Assignment 3 Literature Review; Assignment 4 Requirements Matrix and Scope Statement; Assignment 5 System Design; technical design document (技术方案.md); app/ and server/ README files (v6, Oct 20, 2026).
- Source code v6: app/app.py, config.py, settings.py, pointing.py, tracker.py, radar.py, vision.py, objects.py, hw.py, mqtt_client.py, notify.py, metrics.py, tools/wifi_setup.py; server/ (mosquitto.conf, acl, etec306_cloud.py, wa/bot.js).
- Measured results: tools/servo_tests.json (T2, T3, T7, Oct 6, 2026).
- ETEC326 Preliminary Project Plan (revised, Nov 9, 2026). GitHub: https://github.com/Makabakamanihong/ETEC306/tree/main/docs/etec326-milestone1